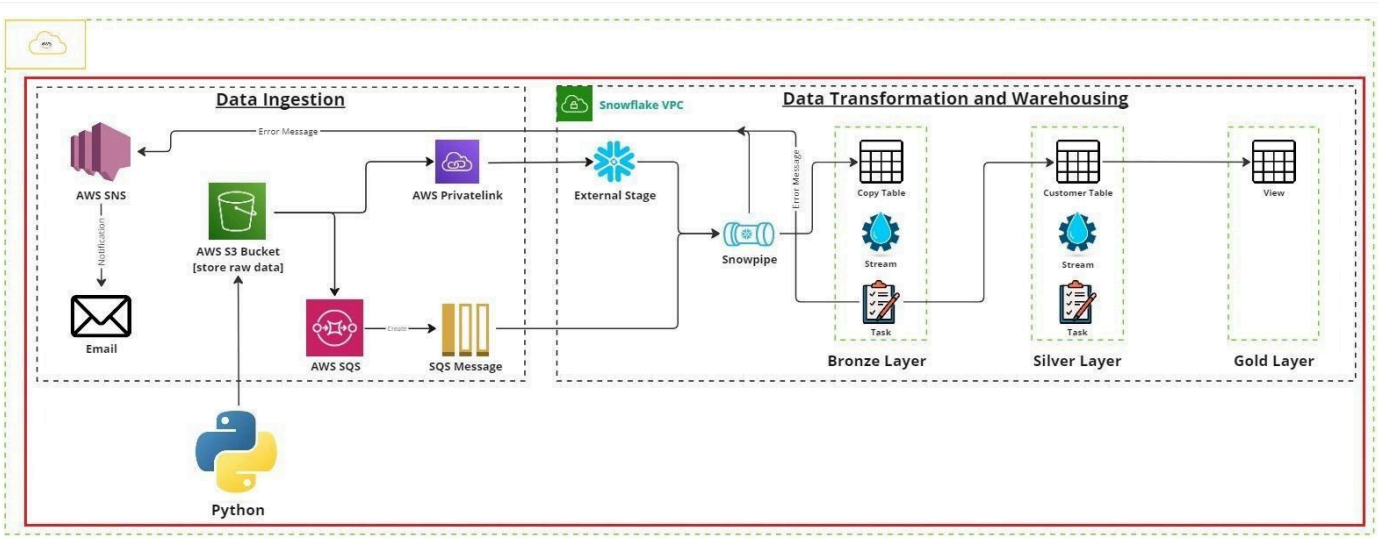
Technical Overview

In this document, we are going to set up a task in snowflake which is going to run a stored procedure containing the SCD1 logic to transform the data in the customer target table

Architecture



Pre-requisites:

- Snowflake Account

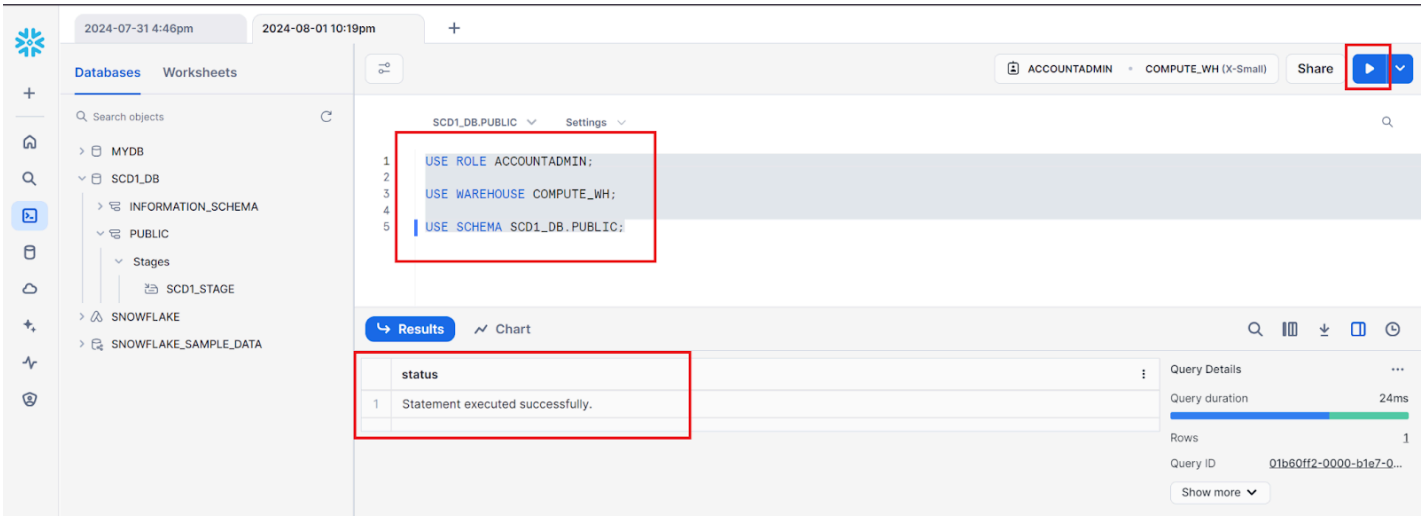
Create a Customer Target table in Snowflake

- Go to the snowflake console and open the already opened SQL worksheet page (if not opened, open a new SQL worksheet) and execute the below SQL statements to select your role,warehouse and schema accordingly by clicking the play button on the top right corner

```
USE ROLE ACCOUNTADMIN;

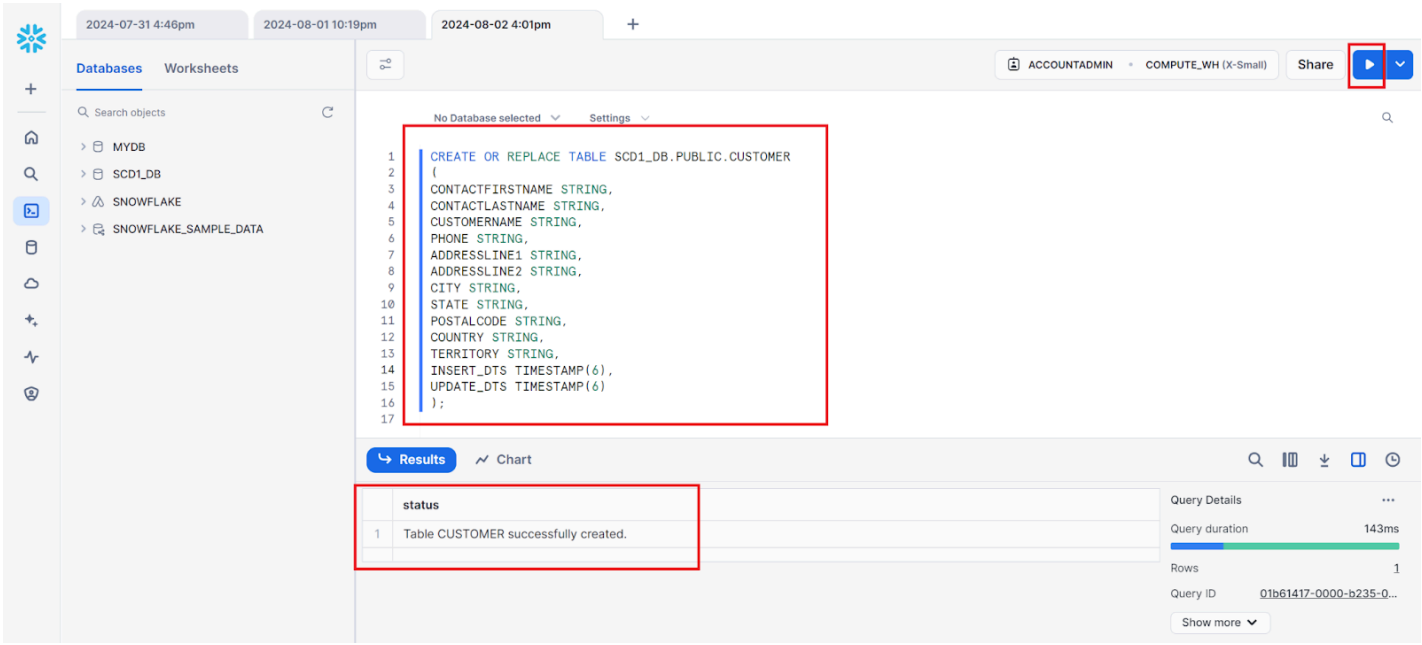
USE WAREHOUSE COMPUTE_WH;

USE SCHEMA SCD1_DB.PUBLIC;
```

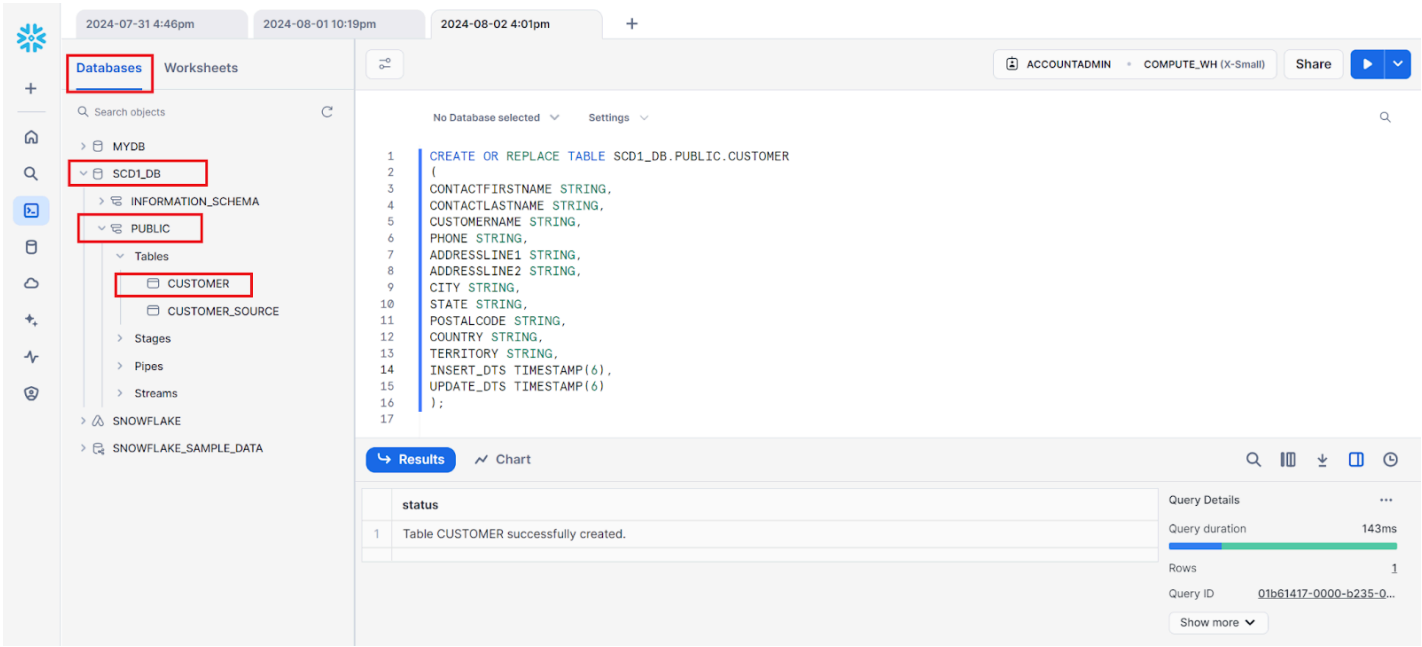


- Execute the below code in the snowflake worksheet to create a customer target table to perform required SCD1 logic using the stored procedure by executing it through a task.

```
CREATE OR REPLACE TABLE SCD1_DB.PUBLIC.CUSTOMER
(
CONTACTFIRSTNAME STRING,
CONTACTLASTNAME STRING,
CUSTOMERNAME STRING,
PHONE STRING,
ADDRESSLINE1 STRING,
ADDRESSLINE2 STRING,
CITY STRING,
STATE STRING,
POSTALCODE STRING,
COUNTRY STRING,
TERRITORY STRING,
INSERT_DTS TIMESTAMP(6),
UPDATE_DTS TIMESTAMP(6)
);
```



- Once the table is created successfully, refresh your page to validate that the table is created under the correct database and schema.



Create a stored procedure to perform SCD1 logic in customer target table

- Execute the below code in the snowflake worksheet to create a stored procedure which is going to perform the below two activity
  - Read the data from the stream created for the customer table and create a work table
  - Load the data into the customer table and perform the required SCD1 logic

CREATE OR REPLACE PROCEDURE SCD1_DB.PUBLIC.CUSTOMER_SP()

```
RETURNS VARCHAR(50)

LANGUAGE JAVASCRIPT

EXECUTE AS CALLER

AS

$$

try {


//Create statement BEGIN, Begins a transaction in the current session


snowflake.execute({sqlText:`BEGIN TRANSACTION;`});


//load data from Customer Stream to a temp table


snowflake.execute({sqlText:`CREATE OR REPLACE TEMPORARY TABLE
SCD1_DB.PUBLIC.CUSTOMER_TEMP

AS

SELECT

CONTACTFIRSTNAME,

CONTACTLASTNAME,

CUSTOMERNAME,

PHONE,

ADDRESSLINE1,

ADDRESSLINE2,

CITY,

STATE,

POSTALCODE,

COUNTRY,

TERRITORY,

CURRENT_TIMESTAMP(6) AS INSERT_DTS,

CURRENT_TIMESTAMP(6) AS UPDATE_DTS

FROM
```

```
SCD1_DB.PUBLIC.CUSTOMER_STREAM;`});

//Perfom the required SCD1 logic on the Customer Target table based on the primary column

snowflake.execute({sqlText:`MERGE INTO SCD1_DB.PUBLIC.CUSTOMER TGT
USING SCD1_DB.PUBLIC.CUSTOMER_TEMP TMP
ON TGT.CONTACTFIRSTNAME = TMP.CONTACTFIRSTNAME
AND TGT.CONTACTLASTNAME = TMP.CONTACTLASTNAME

WHEN MATCHED THEN UPDATE SET
TGT.CUSTOMERNAME = TMP.CUSTOMERNAME,
TGT.PHONE = TMP.PHONE,
TGT.ADDRESSLINE1 = TMP.ADDRESSLINE1,
TGT.ADDRESSLINE2 = TMP.ADDRESSLINE2,
TGT.CITY = TMP.CITY,
TGT.STATE = TMP.STATE,
TGT.POSTALCODE = TMP.POSTALCODE,
TGT.COUNTRY = TMP.COUNTRY,
TGT.TERRITORY = TMP.TERRITORY,
TGT.UPDATE_DTS = TMP.UPDATE_DTS

WHEN NOT MATCHED THEN INSERT
(
CONTACTFIRSTNAME,
CONTACTLASTNAME,
CUSTOMERNAME,
PHONE,
ADDRESSLINE1,
ADDRESSLINE2,
CITY,
STATE,
```

```
POSTALCODE,
COUNTRY,
TERRITORY,
INSERT_DTS,
UPDATE_DTS
)
VALUES
(
TMP.CONTACTFIRSTNAME,
TMP.CONTACTLASTNAME,
TMP.CUSTOMERNAME,
TMP.PHONE,
TMP.ADDRESSLINE1,
TMP.ADDRESSLINE2,
TMP.CITY,
TMP.STATE,
TMP.POSTALCODE,
TMP.COUNTRY,
TMP.TERRITORY,
TMP.INSERT_DTS,
TMP.UPDATE_DTS
);`});

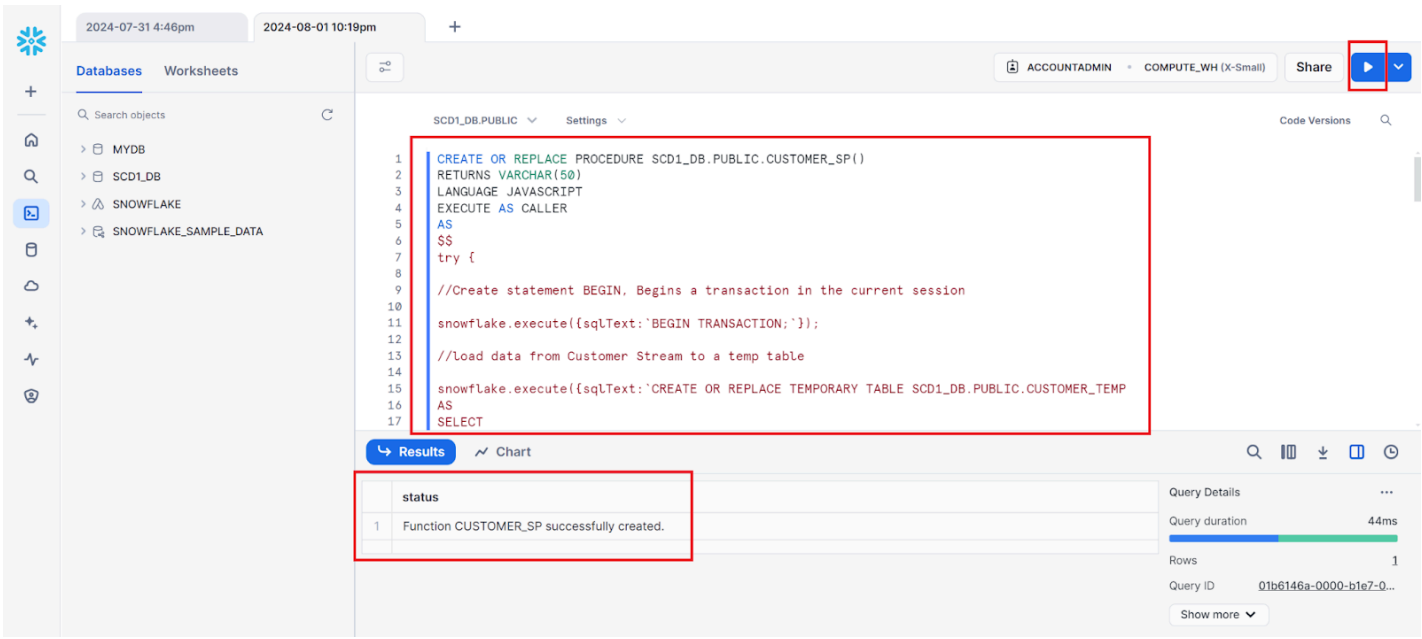
//Create statement COMMIT, Commits an open transaction in the current session

snowflake.execute({sqlText:`COMMIT;`});

//Statement returned for info and debuging purposes

return "Store Procedure Executed Successfully";
}
```

```
catch (err)
{
    result = 'Error: ' + err;
    snowflake.execute({sqlText:`ROLLBACK;`});
    throw result;
}
$$;
```

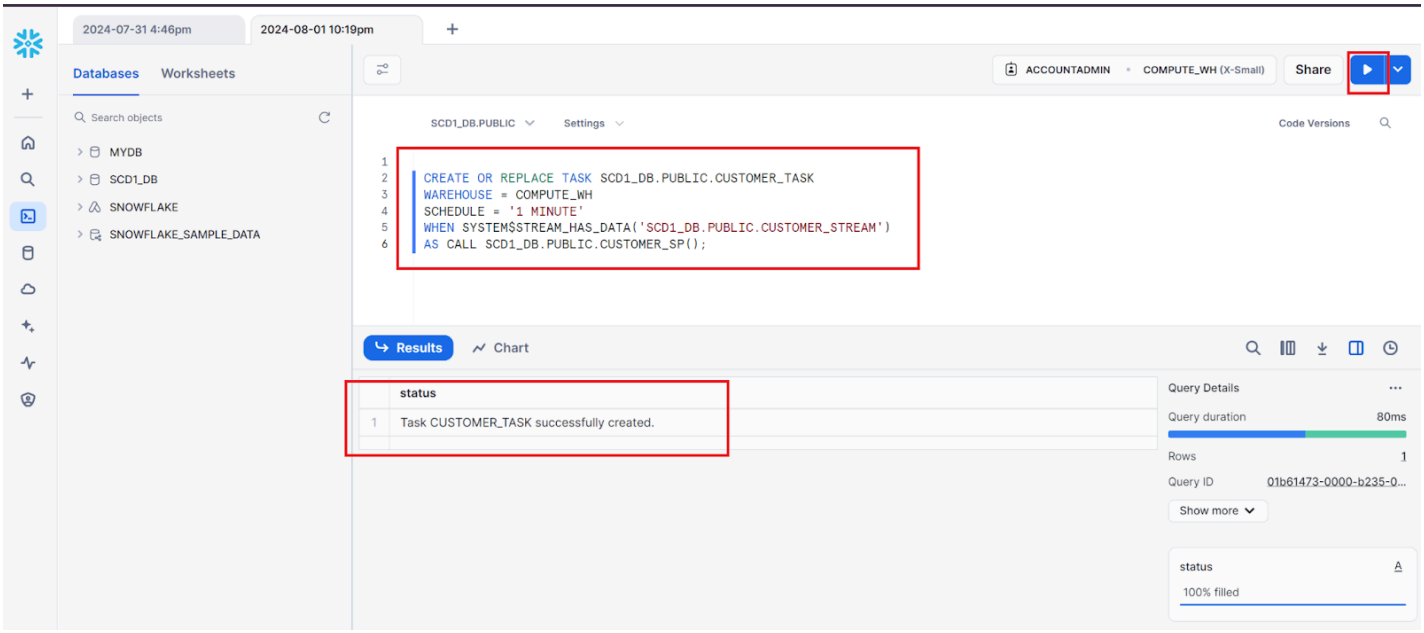


Create a snow task to execute the above stored procedure

- Execute the below code in the snowflake worksheet to create a snow task to execute the above created stored procedure every 1 min

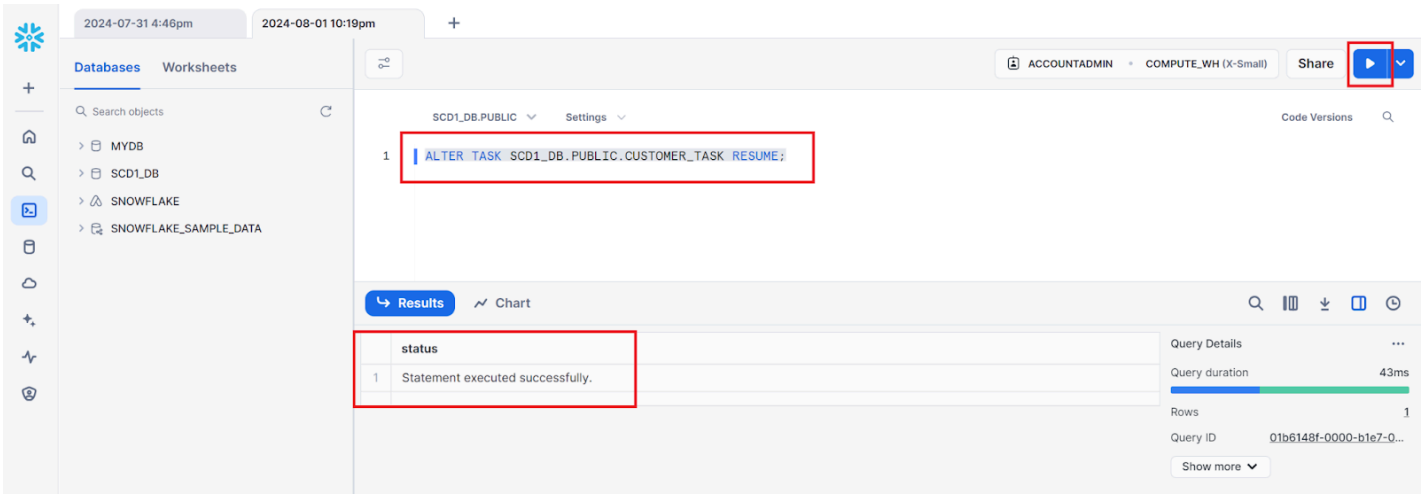
```
CREATE OR REPLACE TASK SCD1_DB.PUBLIC.CUSTOMER_TASK
WAREHOUSE = COMPUTE_WH
SCHEDULE = '1 MINUTE'
WHEN SYSTEM$STREAM_HAS_DATA('SCD1_DB.PUBLIC.CUSTOMER_STREAM')
AS CALL SCD1_DB.PUBLIC.CUSTOMER_SP();
```

.



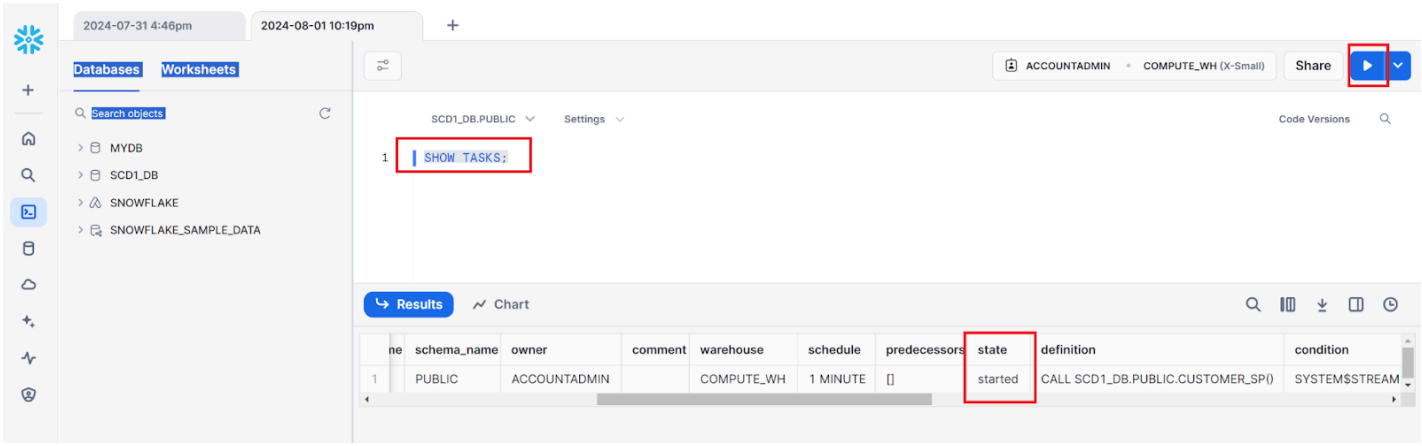
- Alter the above created snowflake task to RESUME by running the following query in the SQL worksheet

ALTER TASK SCD1_DB.PUBLIC.CUSTOMER_TASK RESUME;



- Verify if the task is now in the STARTED state by running the below SQL query in the snowflake worksheet

SHOW TASKS;



VOILA! YOUR SUCCESSFULLY CREATED THE SNOWFLAKE TASK TO EXECUTE STORED PROCEDURE CONTAINING SCD1 LOGIC

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